LABORATORY REPORT

Name: Mar Julius Ubas

Course & Section: IT / WS101

Lab 8

Date: May 7,, 2026

I. OBJECTIVES

Integrate a Spring Boot REST API with a MySQL database using Spring Data JPA.

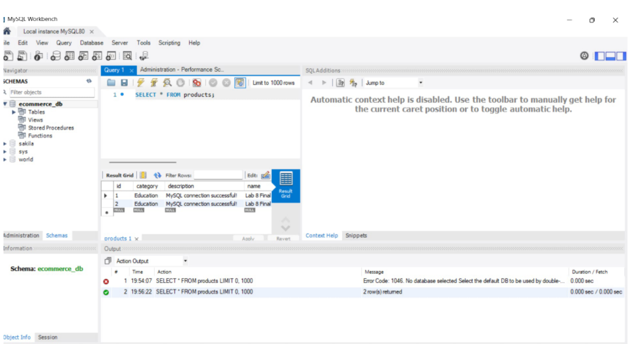
Create and configure database-backed API endpoints.

Consume RESTful web services from the frontend using the JavaScript Fetch API.

II. DOCUMENTATION OF RESULTS

1. Database Schema (MySQL Workbench)

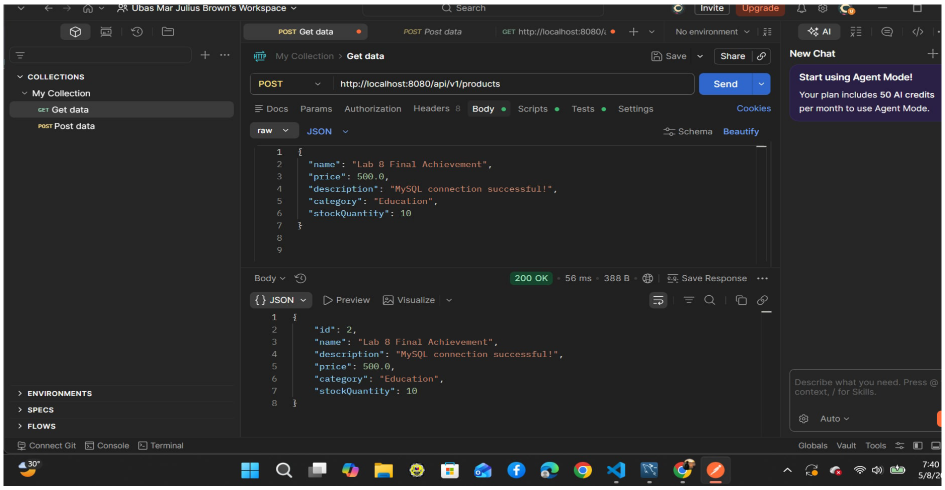
Description: This screenshot shows the successful creation and population of the database table in MySQL Workbench. The backend application connects to this table to perform CRUD operations.



Edit with WPS Office

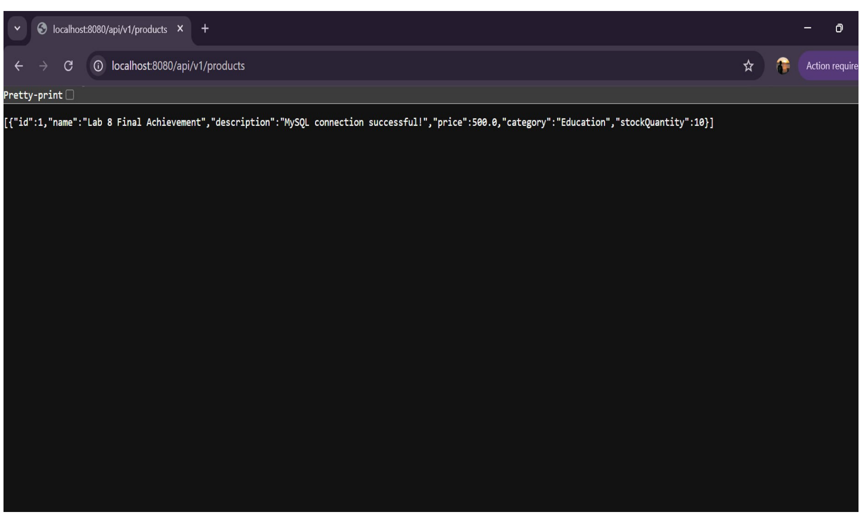
2. API Verification (Postman)

Description: A GET request was sent to the local server via Postman to verify that the backend correctly retrieves data from the database and returns it in JSON format.



3. Frontend Integration (Web Browser)

Description: The final web interface where the JavaScript Fetch API is used to asynchronously request data from the Spring Boot backend and display it dynamically in the browser without a page reload.



Edit with WPS Office

III. ANALYSIS / CONCLUSION

In this laboratory activity, I successfully implemented a full-stack connection between a MySQL database, a Spring Boot backend, and a JavaScript frontend. I learned how Spring Data JPA simplifies database interactions and how the Fetch API allows for dynamic content updates on the client side. By resolving CORS issues and handling promises in JavaScript, I was able to display live database records on a web page, which is a fundamental skill in modern web development

Edit with WPS Office